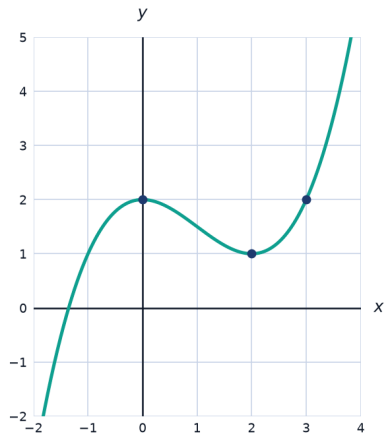
Linear inequalities (Extended)

Core Concepts: Linear inequalities (Extended)

1 For the model $f(x) = 6x + 6$ in linear inequalities (extended), evaluate $f(2)$.



a. 18

b. 24

c. 12

d. 22

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 For the model $f(x) = 6x + 6$ in linear inequalities (extended), evaluate $f(3)$.

a. 24

b. 30

c. 18

d. 28

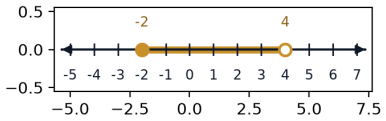
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 6x + 6$ in linear inequalities (extended), evaluate $f(4)$.

a. 30

b. 36

c. 24

d. 34

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

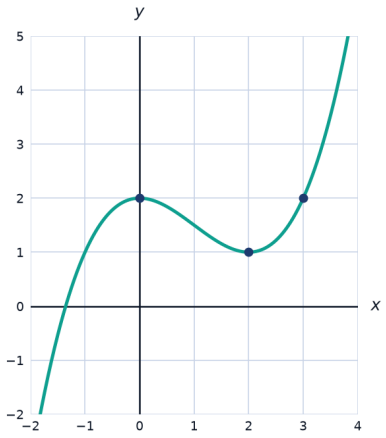
9 A data set recorded for linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5



Exam-Style Problem Solving & Modeling

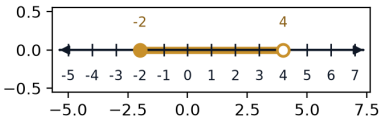
10 For the model $f(x) = 6x + 6$ in linear inequalities (extended), evaluate $f(5)$.

a. 36

b. 42

c. 30

d. 40



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for linear inequalities (extended) consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647